



Paw Notes (Mac Desktop Application)

Software Architecture & Technical Design Specification

Target OS: macOS

Stack: Tauri 2.0 + React

Theme: Cozy Cartoonish Puppy

Use Case: German Class Notes

1. UX & UI Design Concept

Paw Notes is built as a warm, cozy "digital desk" note-taking tool tailored for Mac desktop workflows. It prioritizes pleasant visual design and quick typing with handwriting typography rather than complex trackpad freehand drawing.



Visual & Aesthetic Theme

- **Palettes:** Pastel cream (#FAF6EE), soft yellow sticky notes (#FFF3B0), sage green accents, warm brown borders.
- **Themes:** Golden Retriever, Corgi, and Shiba Inu color schemes.
- **Mascots:** Cute puppy empty states, success badges, and decorative stickers.



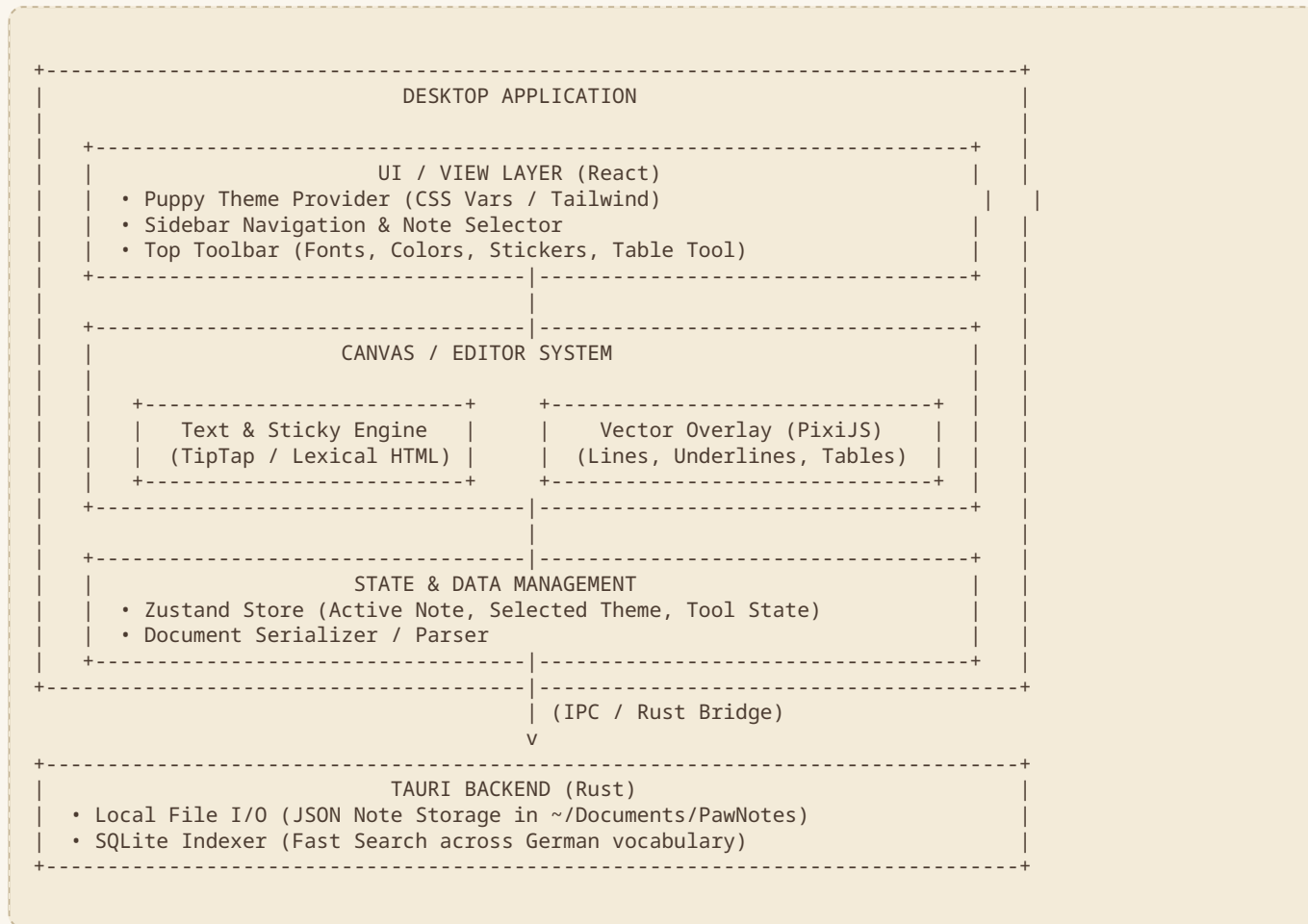
Typography & Tools

- **Handwriting Fonts:** Caveat, Indie Flower, or Patrick Hand.
- **UI Fonts:** Rounded sans-serifs (Nunito / Fredoka).
- **Vector Overlay:** Straight lines, underlines, translucent highlighters, and vocabulary table generators.

2. Core Feature Set (V1 Scope)

- **Handwriting Style Rich Text:** Instant font family switching between standard clean sans-serif and casual handwriting fonts. Customizable notebook background overlays (lines, grid, dot paper).
- **Precision Lines & Tables:** Vector straight line tool for quick underlines or linking concepts. One-click vocabulary table generator (e.g., *Artikel* / *Wort* / *Bedeutung*).
- **Interactive Sticky Notes & Stickers:** Draggable, floating sticky notes for grammar rules and drag-and-drop puppy stickers to mark progress.
- **Fast Local Search:** Instant full-text search across all German vocabulary and grammar notes.

3. System Architecture



4. Recommended Tech Stack

Layer	Technology	Rationale
Desktop Wrapper	Tauri 2.0 (Rust)	Ultra-lightweight RAM footprint (~20MB) vs Electron (~500MB). Native macOS performance.
Frontend	React + TypeScript	Robust component model for rich toolbars, sidebars, and reactive theme engines.
Rich Text Engine	TipTap (ProseMirror)	Highly customizable editor with support for custom fonts, tables, and highlighters.
Vector Layer	SVG / PixiJS Canvas	Renders crisp straight lines, underlines, arrow links, and interactive stickers.
State & Styling	Zustand + Tailwind CSS	Fast state management and CSS variable theme switching (Golden Pup, Corgi, Shiba).
Data Persistence	Local JSON + SQLite	JSON files store document structures; SQLite powers instant search indexing.

5. Document Schema Specification

```
{
  "id": "note_de_vocab_01",
  "title": "Kapitel 2 - German Noun Genders",
  "createdAt": "2026-08-14T08:30:00Z",
  "theme": "golden_retriever",
  "backgroundStyle": "lined",
  "fontFamily": "Caveat",
  "content": {
    "textHtml": "
```

Kapitel 2: Im Supermarkt

Nouns and vocabulary rules...

```
",
  "stickyNotes": [
    {
      "id": "sticky_1",
      "x": 450,
      "y": 120,
      "color": "#FFF3B0",
      "text": "Rule: Words ending in -ung are ALWAYS 'die'!"
    }
  ],
  "vectors": [
    {
      "type": "straight_line",
      "startX": 50,
      "startY": 90,
      "endX": 300,
      "endY": 90,
      "strokeColor": "#FF8C42",
      "strokeWidth": 3
    }
  ],
  "stickers": [
    {
      "id": "stk_1",
      "type": "puppy_happy",
      "x": 380,
      "y": 20,
      "scale": 1.0
    }
  ]
}
}
```